

1. Answer any two questions: (25X2=50)

a) How does Carbon Monoxide (CO) affect human health? What is meant by saturation concentration of CO? Differentiate between primary air pollutants and secondary air pollutants. Derive the basic equation for dry adiabatic lapse rate showing its variation with height. Why is Conditional stability so named? What is meant by a stable atmosphere? Explain the role of thermal inversion behind occurrence of high air pollution during the winter periods. (4+3+5+5+2+3+3=25)

b) Derive the wave equation? Derive the expressions for sound intensity level, sound pressure level and sound power level. Establish the relationship between the three. (8+9+8=25)

c) Enumerate the basic Gaussian Plume Dispersion Equation clearly defining all the terms involved in the expression. Modify the equation as per the following conditions: i) the receptor is at ground level with ground reflection having taken place ii) both the source and the receptor are at ground level with no ground reflection iii) plume centreline ground level concentration with no ground reflection. What are the assumptions behind the basic Gaussian Plume Dispersion Model? Derive the followings $\sigma_z = 0.707H$ and $\sigma_z = 0.47H$, where the notations have their usual meanings. (4+6+3+6+6=25)

d) What is meant by Criteria Air Pollutants? What is an Air Quality Index and how does it differ from Exceedance Factor? Explain the difference between the earlier way of representing Air Quality Index and the present way of reporting the same? What are the commonly used noise pollution control measures. What is meant by threshold limit of audibility and threshold limit of pain (4+5+6+5+5=25)

2. Answer the following questions: (15X2=30)

i) It is estimated that 100 gm/sec of sulphur dioxide is being emitted from a petroleum refinery from an average effective stack height of 150 m. At 10 am on an overcast winter morning with a surface wind speed of 9m/sec, what will be the ground level concentration directly downwind from the refinery at a distance of 1200 m. Assume $\sigma_y = 60m$ and $\sigma_z = 30m$. (7.5)

Considering ground reflection

ii) A burning dump emits 9.5 gm/sec of oxides of nitrogen. Find out the concentration of oxides of nitrogen averaged over approximately 25 mins, from this source directly downwind at a distance of 5 km on an overcast night with a wind speed of 12m/sec. Consider the dump to be a point ground-level source with no effective rise. Assume $\sigma_y = 350m$ and $\sigma_z = 90m$ respectively. (7.5)

b) Traffic Noise data are shown in the table below: ((15)

Time (s)	Sound Pressure Level dB(A)
10	74
20	76
30	77
40	70
50	82
60	86
70	78
80	74
90	75
100	72

Compute L_{eq}

ALIAH UNIVERSITY
End-Term Examination- 7th Semester
Session-2022-2023

Year: 4th Year
Semester: 7th Semester
Subject: Irrigation Engineering
Subject Code: CENUGPE04

Full Marks: 80
Time: 3hr.

Answer question no.1 and any three from the rest:

1. Write short notes on the following:

[4*5=20]

- ~~i)~~ Drip irrigation method.
- ii) Types of weir according to cross section.
- ~~iii)~~ Border strips and sprinkler methods of irrigation fields.
- ~~iv)~~ Water-shed canals and contour canals.
- ~~v)~~ Distribution system for canal irrigation.

2. a) What is meant by surface and sub-surface irrigation; and what are their types? Discuss briefly the various techniques used for distributing water in the farms. [2+4+4=10]

b) What is meant by 'Border flooding'?

c) Determine the time required to irrigate a strip of land of 0.04 hectares in area from a tube-well with a discharge of 0.02 cumec. The infiltration capacity of the soil may be taken as 5 cm/hr and the average depth of flow on the field as 10 cm. [2+8=10]

3. a) i) What is meant by 'Duty' and 'Delta' of canal water? What are the factors affecting duty? Derive a relationship between duty and delta for a given base period. [3+3+4=10]

b) Find the delta for sugarcane when its duty 730 hectares /cumec on the field and the base period of the crop being 110 days. [5]

c) Water is released at the rate of 5 cumec at the head sluice. If the duty at the field is 100 hectares / cumec and the loss of water in transit is 30%, find the area of the land that can be irrigated. [5]

4. a) i) Differentiate between 'Alluvial' and 'Non-Alluvial' canals. ii) What are different ways in which the irrigation canals can be aligned? iii) Define culturable command area and intensity of irrigation. [2+5+3=10]

b) i) Write the purposes of diversion head work. ii) Differentiate between a weir and a barrage. [5+5=10]

5. a) What are the main component of head work? What is afflux? [6+2=8]

b) i) What is meant by gravity dams? ii) Draw a net sketch of gravity dams showing different component of a gravity dam. iii) Explain the various forces acting on a gravity dam. [2+3+7=12]

Odd (Autumn) Semester Examination 2022
Paper Code: MBAUGHU01; Paper name: Industrial Economics & Management
B.Tech (CEN) VIIth Semester
Full Marks: 80; Time: 3Hrs.

*(The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as possible)*

GROUP: A (Answer all the questions)

(1 x 10 = 10)

1.

① Which of the following is true about managers within an organisation?
(a) Management skills apply to managers at top levels in an organisation
☒ (b) Management skills apply to managers at all levels in an organisation
(c) Management skills apply to managers at middle levels in an organisation
(d) Management skills apply to managers at executive levels in an organisation

② Management is _____
(a) an art (b) a science ☒ (c) an art and a science (d) an art not science

③ GDP stands for
(a) Gross Development Policy (b) Generic Development Policy
☒ (c) Gross Domestic Product (d) Generic Domestic Product

IV. Which of the following comes first?
☒ (a) Recruitment (b) Selection (c) Training (d) Appraisal

V. This function of Management related to placing the right person at the right job is
(a) Organising ☒ (b) Staffing (c) Planning (d) Controlling

⑥ Top management is concerned with formulation of
(a) guidelines for supervisors ☒ (b) long-term plans
(c) short-term plans (d) None of these

VII. Which of the following is not included in 4Ps of Marketing?
(a) Product (b) Price ☒ (c) People (d) Place

⑧ What is Unity of Command in Principles of Management?
(a) Workers are given a specialized task to do
(b) Necessary to ensure that managerial commands are carried out
(c) Needed within an organization for it to run effectively
☒ (d) An employee should have only one boss

IX. What is Remuneration in Principles of Management?
(a) Employees work in harmony towards the same objective
(b) Interests of the team should take precedence over personal ones
☒ (c) Paid fair wages for the work that they carry out
(d) The concentration of power in the hands of the authority

X. _____ is an obligation to perform certain functions and achieve certain results.
(a) delegation (b) responsibility (c) centralisation (d) decentralisation

GROUP: B (Answer any five questions)

(5 × 5 = 25)

- ✓ 2. Elaborate the functions of Management. ✓
- ✓ 3. What do you understand by 'decentralization'? Why is it preferred than 'centralization' policy any organization?
4. What is the significance of 'span of management' in an organization?
5. Why 'corporate governance' has become inevitable in present time?
6. Write a short note on 'stress management'.
7. Define decision-making and explain the process of decision-making that affects the efficiency of the business decisions.
- ✓ 8. Describe 'Maslow's hierarchy of need'. ✓

GROUP: C (Answer any three questions)

(15 × 3 = 45)

9. What do you understand by 'marketing'? Elaborate the need of marketing.
10. Explain the importance of 'logistic and supply chain management' in an organization.
11. Why recruitment is considered as a positive process and selection as a negative process? Explain the steps of selection process.
12. What is 'break-even point' for any start-up? Suppose you have just launched a start-up company and want to reach at break-even point within a year; show the different components of your strategy quantitatively.
13. Write short note: (a) Kaizen (b) Economic Order Quantity (EOQ)

Odd(Autumn) Semester Examination 2022
Paper Code: MBAUGOE01; Paper name: Entrepreneurship Development
Bachelor of Technology Civil Engineering VIIth Semester
Full Marks: 80; Time: 3Hrs.

(The figures in the margin indicate full marks.)
Candidates are required to give their answers in their own words as far as possible)

GROUP: A (Answer all the questions) (1 x 10 = 10)

- I. Who should be involved in preparing a firm's business plan?
(a) Accountant (b) Engineer ☒ (c) Entrepreneur (d) None of the above
- II. The process of creating something new is called.
(a) Management (b) Business ☒ (c) Innovation (d) None of the above.
- III. What is the purpose of a feasibility study before starting a new venture?
(a) Estimate sales (b) Exploring for potential customers
☒ (c) To understand if there are any barriers to success (d) none of the above
- IV. An entrepreneur who is the owner of more than one business is called:
☒ (a) Corporate entrepreneur (b) Intrapreneur
☒ (c) Portfolio entrepreneur (d) none of the above
- V. Which of these is not a category of external forces that affects a business?
(a) Competitive forces (b) Technological forces
(c) Economic forces (d) Socio- economic forces
- VI. Ownership position of less than 50% in a business is known as:
☒ (a) Majority interest (b) Joint venture (c) Minority interest (d) none of the above
- VII. If an entrepreneur takes a decision on behalf of their enterprise, it is known as....
(a) Routine decision (b) Strategic decision
(c) Personal decision ☒ (d) Organizational decision
- VIII. District industries centers are located
(a) In each district (b) in each state ☒ (c) Only in selected districts (d) only in selected states
- IX. is the first development bank of the country.
(a) ICICI ☒ (b) IDBI (c) SFC (d) IFCI
- X. MSME is defined under which Act?

(a) MSMED Act 2006 (b) MSMED Act 2001 (c) MSMED Act 1999 (d) MSMED Act 2004

GROUP: B (Answer any five questions)

(5 x 5 = 25)

2. Discuss the characteristics of a successful entrepreneur.
3. Write short notes on venture capital.
4. What are the problems faced by women entrepreneurs in India?
5. Differentiate between Entrepreneur and Manager.
6. Explain the concept of Corporate Social Responsibility in business.
7. What is SWOT analysis?
8. Explain the types of entrepreneurs.

GROUP: C (Answer any three questions)

(15 x 3 = 45)

9. Suppose you are planning to start a new venture after graduating from the university. Explain in details all the footsteps in setting up the new business enterprise.
10. Explain the role of MSME in the development of Indian Economy.
11. Discuss Entrepreneurship as a Career. What are the factors favoring entrepreneurship a career option?
12. "Entrepreneurs are born not made"- comment on this statement
13. Explain the features, advantages and disadvantages of proprietorship, partnership, companies and cooperatives.
